

[Manan Sonali Kher]

[Computer Engineering]

Company: WorkIndia

Job Profile: SDE

Offer Type: [Internship + Placement]

Internship Stipend: 40,000

CTC: 15 + 1 LPA

Location: Bangalore

Process Date: 09/07/2025

Placement Process:

There were 5 rounds in total, all conducted in a single day. Try to prepare yourself on API development, either Django or NodeJS/ExpressJS. Also practice SQL queries from SQL 50, and learn the basics of designing schemas. Also, all the interviews were conducted online, and we had to sit in the panel room allocated to us, and join the interview meet link on our laptop.

Resume Shortlisting: Shortlisting was done based on the resumes that we had submitted on Superset when we applied to WorkIndia. Also, while applying, in the form they had asked whether I had plans to go for Masters later on. To this, I said NO. For the resume, follow the format of Jake's resume, and try to keep it 1 page. Properly include your experience, projects and skills, and make sure you can speak about anything that you put in your resume.

[Jake's Resume Format](#)

 MananKher__Resume.pdf

1. **Rapid Fire Round** [72 candidates appeared for this round]

This was the first round, and we were all asked to arrive by 8:25 am, and were asked to assemble in 103 by 8:50 am. We thought that we would first receive instructions on how the rapid fire round would be like. But at 9 am sharp, we all receive an email, containing a google form link, and one of the WorkIndia coordinators comes into the room, and tells us to start the test immediately, and that the form will stop taking submissions after 9:15 am. So make sure you have your wifi and laptop battery charging setup. Luckily I was already connected to the wifi and had my laptop charged, so I was able to start the test immediately.

It was an MCQ test which contained basic technical questions on Java, Python and C++, like answering the time complexity, space complexity and output of code snippets. Some questions on basic OS and CCN like deadlocks and DNS, some github commands like listing all branches, stash and pop. Some questions on prefix and postfix increment operators, and some basic data structure questions like which data structure is LIFO, FIFO. The objective is to answer as many questions as you can within 15 min. It contained around 40 to 50 questions. I was able to answer all of them, some with a little uncertainty, and submit.

2. **Debugging Round** [46 candidates appeared for this round]

For this round, we had to join the meet link on our laptop. The interviewer then asked whether I prefer Django or NodeJs. I chose Django. Then he sent a link to a small existing django codebase containing an Attendance Tracker project, with a README file containing instructions on what was to be performed. Initially, while starting the server, there were 2 bugs that were required to be fixed.

One bug was in the models.py, a syntax error in the choices parameter passed to a models.CharField.

Expected syntax: `status = models.CharField(max_length=20, choices=[('active', 'Status active'), ('inactive', 'Status inactive')])`

Given syntax: `status = models.CharField(max_length=20, choices=['active', 'inactive'])`

The second bug was in the views.py, another syntax error in which the key of a python dictionary was being accessed incorrectly.

Expected syntax: `[s["name"] for s in students]`

Given syntax: `[s.name for s in students]`

After fixing the errors, I started the server. The server was running now. After this, I was asked to implement an API endpoint to mark a student's attendance. For this, I was initially adding an attendance field in the existing Student table. But the interviewer asked me for a better way. So I made a separate table, Attendance, which can contain the student_id as foreign key and a date field, to store when the attendance was marked. Then, there was another constraint that a student cannot mark attendance again on the same date. For this, I used the unique_together model option within a Meta class to make the composite field (student_id, date) unique. The interviewer was happy with this. Then I started implementing the API endpoint, and wrote the url and the handler function. But time was running short, so the interviewer just asked to write a comment inside the handler function on how I would implement the endpoint.

Throughout the round (around 45 min), I kept telling the interviewer my thoughts and the steps I am taking. Also, googling and using stack overflow was allowed, but no LLMs. The interviewer will judge you based on your knowledge of the programming language, the backend framework, and most importantly your ability to ask the right questions and google them, and find out the answers. I had googled the syntax for using choices parameter, the syntax for implementing the unique together and also the date field.

3. **DSA Round** [15 candidates appeared for this round]

The DSA Round was also conducted in online interview form, and we had to write code on google docs. The questions were of leetcode easy and medium difficulty. The interviewer first asked me to introduce myself. Make sure you have a proper introduction ready, which briefly covers your education, your internship experience and high impact projects. Also, create an introduction which aligns with the company's principles.

First question: You are given a positive integer N. Find X such that the sum from 1 to X-1 equals the sum from X+1 to N. For this, I first suggested the brute force algorithm of iterating through all elements from 1 to N, then for each element, running 2 for loops to calculate the 1-X-1 and X+1-N sum, giving a $O(n^2)$ TC. I then optimized it by suggesting a prefix sum approach. First, we calc the total sum using the sum of n natural numbers formula, and then iterate from 1 to N, calc prefix sum, and subtract from total sum. So whenever left and right sum become equal, we return that element X. I wrote this code, then the interviewer then told me to optimize it further. So I suggested a binary search approach, and conditions would be if $\text{leftSum} > \text{rightSum}$, $e = \text{mid} - 1$ and if $\text{leftSum} < \text{rightSum}$, $s = \text{mid} + 1$, else return mid. Also, I was asked to optimize space complexity as well, so I used the same formula to calculate leftSum (1 to X-1 sum), instead of prefix array. I wrote this code, then the interviewer asked to optimize it even further. I took some time, but thought that if I want to optimize from $O(\log n)$, there must be a direct formula. That's when it clicked me, that I can write an equation, and solve it for X.

$\text{leftSum} = \text{rightSum}$

$(x-1)x/2 = \text{totalSum} - \text{leftSum} - x$

$(x-1)x/2 = n(n+1)/2 - (x-1)x/2 - x$

$(x-1)x = n(n+1)/2 - x$

$x^2 - x = n(n+1)/2 - x$

$x^2 = n(n+1)/2$

$X = \text{sqrt}(n(n+1)/2)$

The interviewer then asked, what does this signify. I said that X will be the sqrt of sum of the n natural numbers. He was happy with this, and explained me that for a solution to exist, the sum of the n natural numbers must be a perfect square.

Second Question: <https://leetcode.com/problems/majority-element/description/>

I first suggested a brute force approach by creating a hashmap (dictionary in python) and storing the frequency of all elements. Then iterating through the hashmap, and returning the element whose frequency is greater than $n/2$.

The optimal approach with constant space complexity is Moore's Voting Algorithm.

 Majority Element - Leetcode 169 - Python

During the DSA round, I was asked why I preferred using Python. I explained that I initially used C++ and had solved around 150 questions on LeetCode in it. But later, I switched to Python because its simpler syntax allowed me to focus more on understanding the logic behind each problem, rather than the boilerplate code. This helped me improve my problem-solving skills more effectively.

Tip: If you switch languages for DSA, be clear about your reason, it's not about which language is "better", but what helps you learn and think better.

Third Question: <https://www.geeksforgeeks.org/dsa/find-a-repeating-and-a-missing-number/>

For this, I was only able to suggest the approach with the frequency array / hashmap. The interviewer wanted me to optimize the space complexity. I started thinking in terms of slow and fast pointer, but on dry running, that approach didn't work. Then the interviewer gave me a hint that I can modify the array in-place. I started thinking along those lines, maybe swapping the numbers. But time was up, and I

wasn't able to optimize it. The interviewer then asked me if I had any questions. I asked them about the various microservices they have built, and what different architectures they have used, along with the methodology they take while deciding the appropriate architecture for these services.

Finally, after receiving the answer, I thanked every person on the call, by their first name.

At this point, I felt I might not get selected for the next round, since I couldn't come up with the most optimal solution for the third question. Still, I kept my hopes up and didn't give up. I began preparing for the next round anyway, asking ChatGPT to help me build schemas for applications like a job recruitment platform.

After quite some time, I finally received the link for the next round! The waiting period can be really nerve-wracking, but it's important to stay calm and think positively. Don't be disheartened if your interview doesn't go as expected, it's never truly over until the list of shortlisted candidates for the next round is out.

Bonus Question (asked to one of my friends):

<https://leetcode.com/problems/minimum-size-subarray-sum/description/>

A further variation of this was asked, by allowing negative integers to exist in the array.

4. **Schema Design Round** [7 candidates appeared for this round]

In this round, I was given a problem statement of an Ecommerce Platform on google docs, along with its various requirements. Also, 2 bonus requirements were given of implementing coupons for orders, and a referral system. Initially, I focused on the main requirements, and identified the main entities from them. I wrote down the table names. Then I started building the relations between these tables, with the help of foreign keys, along with the necessary attributes for each of them. While building the schema, I made sure to obey the normal forms like 1NF and mentioned these terms while making the schema. Make sure that you keep properly explaining the interviewer your thought process behind each and every step, and also make sure to ask doubts on things that might not be clear in the problem statement. I had asked some questions, like should I also focus on the merchant's side, or only on the customer's side, and difference between inventory and warehouse, as it was not very clear from the problem statement. Explain the cardinality of each relationship you are creating (one-one, one-many, many-many).

After creating a rough schema, the interviewer then identified certain areas that could be improved, normalized further. I listened thoroughly and made these changes, and explained how this approach would be more effective in preventing data redundancy. I was then asked to write 2 SQL queries on the schema I had built. This included finding the delivery times of the recent orders, along with their delivery locations. I used JOIN, ORDER BY, LIMIT, DATEDIFF function for subtracting the placed_at from the delivered_at. Then I was asked to find the average delivery time, for which I used the aggregate function.

As I had time remaining, I also went ahead and implemented the bonus requirements. The referral table would be a unary, many-many relationship, and contains referred_by and referred_to (both foreign keys to User table)

Finally, the interviewer asked if I had any questions. I asked him what his first six months at WorkIndia were like. He mentioned that he had joined recently, and that while the initial weeks can be hectic, things become smoother as you get familiar with the processes.

I shared my own experience from an internship at a startup, where the first couple of weeks were also hectic for me, since I was learning a new tech stack while working on live features. But once I got used to the codebase, I was able to contribute meaningfully and build exciting features. That sense of solving real-world problems was deeply satisfying. The interviewer appreciated this and mentioned that I would be a good fit for a startup like WorkIndia.

I would advise learning CTEs while solving SQL 50, because they are a powerful way of dividing your SQL query into manageable and easy-to-understand sequential steps.

5. **Culture Round** [4 candidates appeared for this round]

The culture round felt more like a meaningful conversation than an interview. The interviewer was very friendly, and there's truly no need to be nervous. We spoke about my journey, passions, values, and experiences. It was a heart-to-heart.

The questions covered:

- What got you into this field? Why engineering and not any other role?
- Tell me about your journey so far, your internship, and what you learned from your peers.
- Lowest point in your life and academics.
- Your family, your interests, your short- and long-term goals.
- What do you consider as “good work”?

I answered honestly, sharing how I got into tech through PC gaming and modding, inspired by my grandfather's tech-savviness. I spoke about learning QBASIC in school, then Java, and how my love for coding naturally led me to engineering, where I could solve real-world problems and innovate.

I reflected on my regret of not joining any committee in first year, which limited my exposure, but after second year, I made it a point to step out of my comfort zone: applying for internships, connecting with people on LinkedIn, and seeking out opportunities.

I also shared my internship experience, the meaningful features I built, and how much I learned from my mentors and colleagues. I opened up about my family's support, and my goal to help them by becoming financially independent and making them proud.

When asked about what “good work” means to me, I shared how participating in hackathons changed my perspective. Earlier, good work meant building something cool solo using a new tech. But through hackathons, I realized that true good work is about team collaboration, communication, and building something together that no one could have done alone. It's about leveraging each person's strengths, validating each step as a team, and delivering something meaningful as one unit.

Finally, I asked the interviewer:

- How did WorkIndia start, and what impact were you hoping to create?
- How do you deal with scam jobs on the platform?
- From the core values in the WorkIndia PPT, which one do you consider most important?

It was a genuine and fulfilling conversation where I could truly be myself.

Tip: Don't prepare scripted answers. Reflect on your journey, speak from the heart, and be honest. This round isn't about checking boxes, it's about understanding who you are and what drives you. Also, have a good idea about the companies that you have interned at. The interviewer might ask certain questions like when was the company founded, or how old was your mentor in the company, not to grill you, but to verify authenticity and see how well you know your own experiences. Be honest, and don't try to bluff. It's totally okay to say "I'm not sure" rather than guess.

2 students were offered the role. It was a long day, and the results weren't announced immediately. They got announced after we went back home.

Useful Tips:

1. **Craft a solid self-introduction** — it's your first impression, so make it count.
2. **Take notes during the pre-placement talk** — they can help you frame personalized questions during the interviews.
3. **Practice a backend framework** like Django or Express — learn to set up a server, design schemas, and build basic APIs beforehand. I had done a Django course on Udemy - [Django Masterclass : Build 9 Real World Django Projects](#)
4. I personally did [NeetCode 150](#), then practiced **LeetCode daily** and **previous contest questions**. Focus on mastering **fundamentals and recognizing patterns**, not chasing streaks or ratings. You can also do Striver's A2Z DSA sheet.
5. Use **Indiabix** and **GFG** to revise fundamentals of DBMS, OS, OOPs and CCN. DBMS is the main priority, as the interviews revolved around being able to apply the concepts you learned in DBMS. Just some basic knowledge of OS, OOPS and CCN was required for the rapid fire round.
6. [LeetCode's Top 50 SQL Questions](#) are more than enough to get a strong grip on SQL basics and variations.
7. **Don't panic in interviews** — it's okay to ask for help; interviewers are usually very supportive. **Think aloud throughout each interview** — it helps the interviewer follow your approach and gives them a chance to guide you if needed.

Feel free to contact me on LinkedIn - <https://www.linkedin.com/in/manankher/>